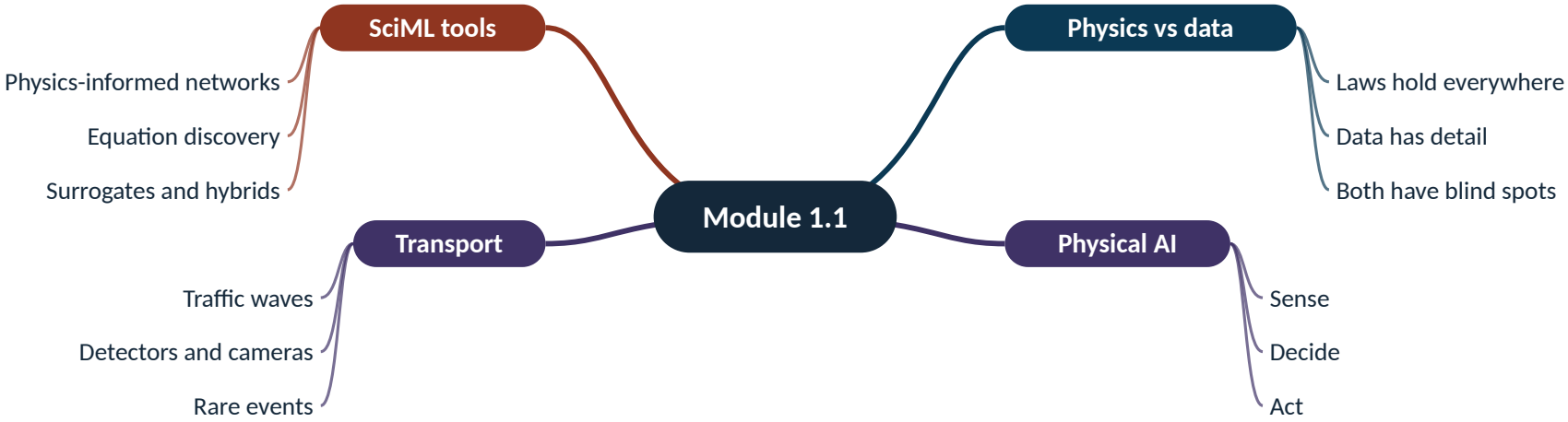
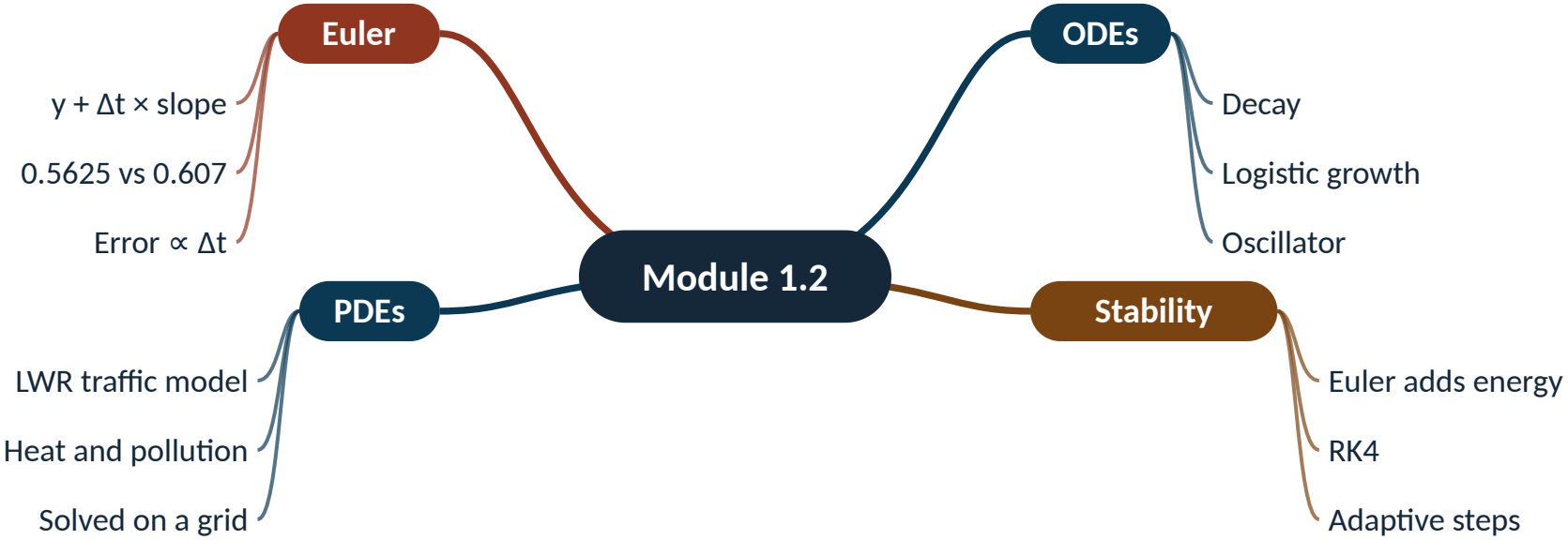
Go 1 mind map


Go 1
1.1 What Physical AI and scientific machine learning are
Physics vs data
SciML tools
Physical AI
Transport
1.2 Differential equations: the language of physics
ODEs
Euler
Stability
PDEs
1.3 Neural networks from scratch
Neuron
Universal approximation
Limits
1.4 Learning as optimisation: loss, gradients, autodiff
Loss
Gradient descent
Better optimisers
Autodiff
1.5 Physics-informed learning: the core idea
Loss
Fewer data
Balance
PINNs
1.6 Data, noise and overfitting
Signal and noise
Training vs test
Defences
1.7 Surrogates and emulators: fast stand-ins
Pay-off
RBF surrogate
Design
1.8 Physical AI: sense, estimate, decide, act
Loop
Kalman filter
Digital twin
Safety

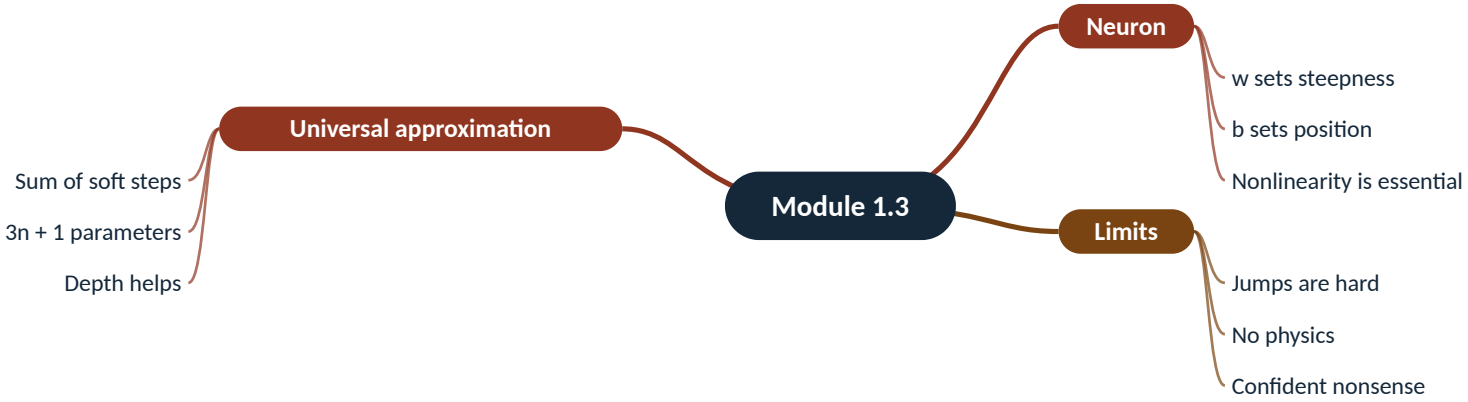
1.1 What Physical AI and scientific machine learning are



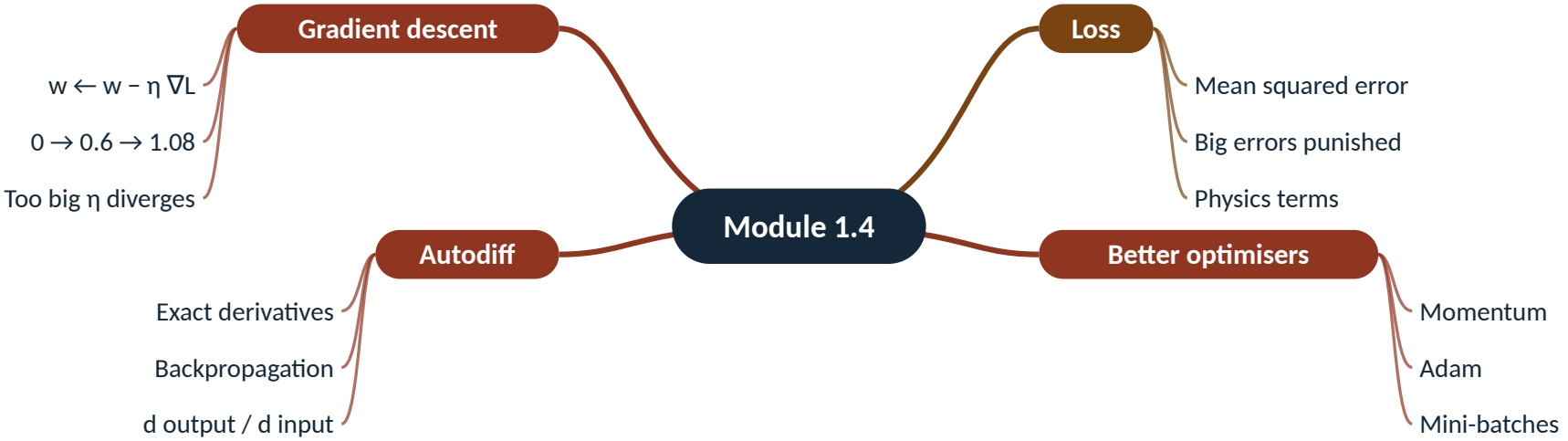
1.2 Differential equations: the language of physics



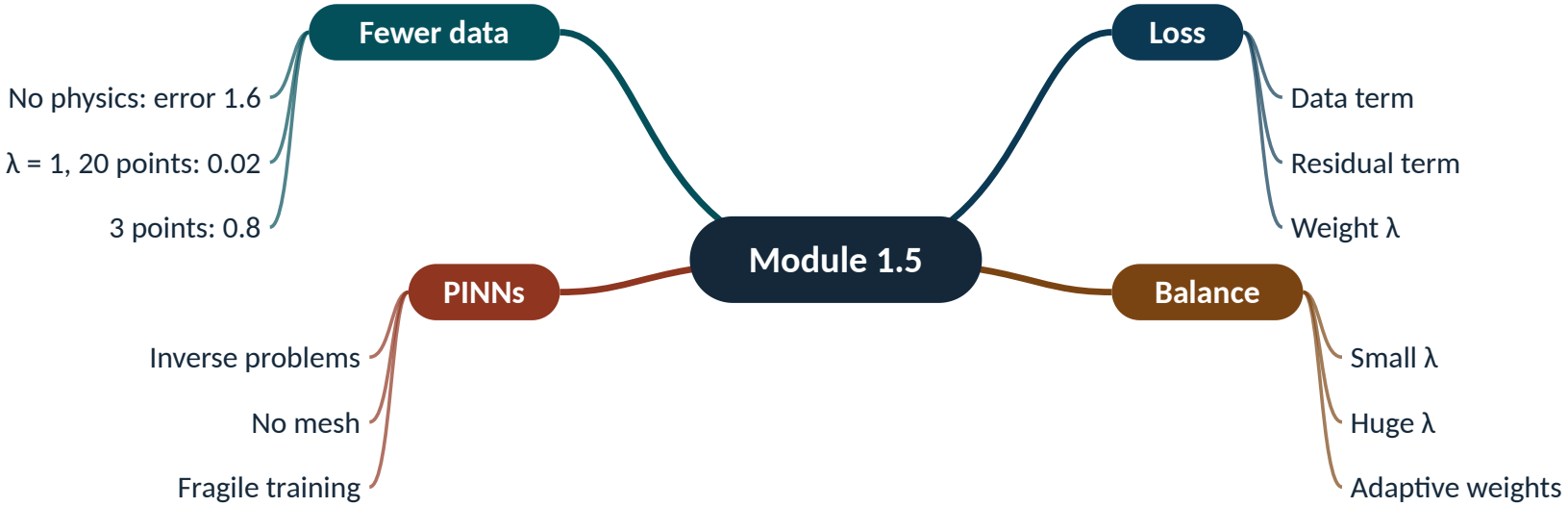
1.3 Neural networks from scratch



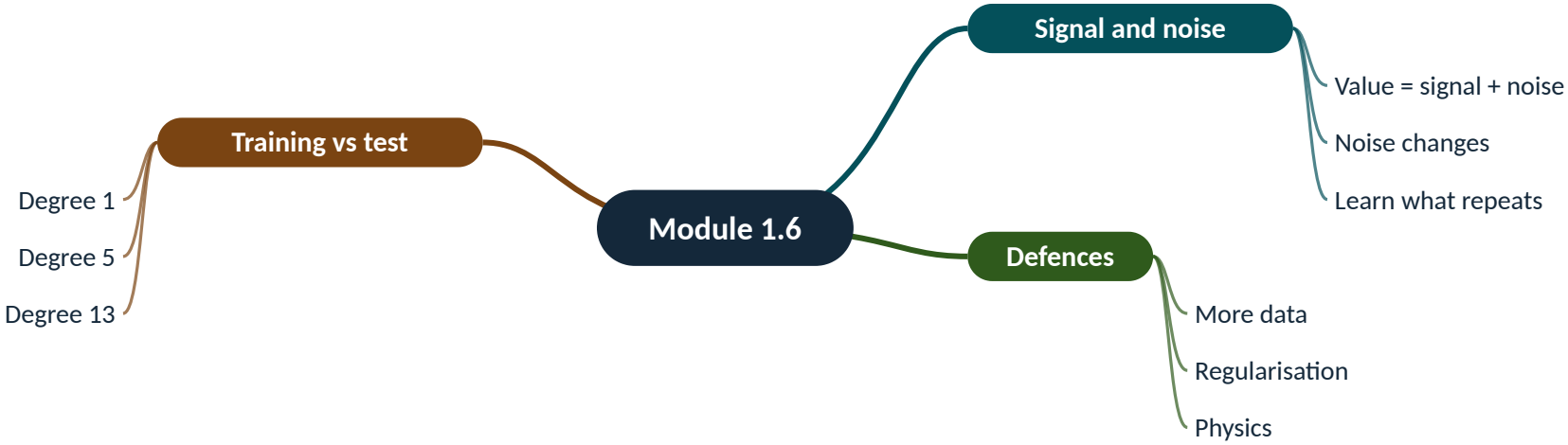
1.4 Learning as optimisation: loss, gradients, autodiff



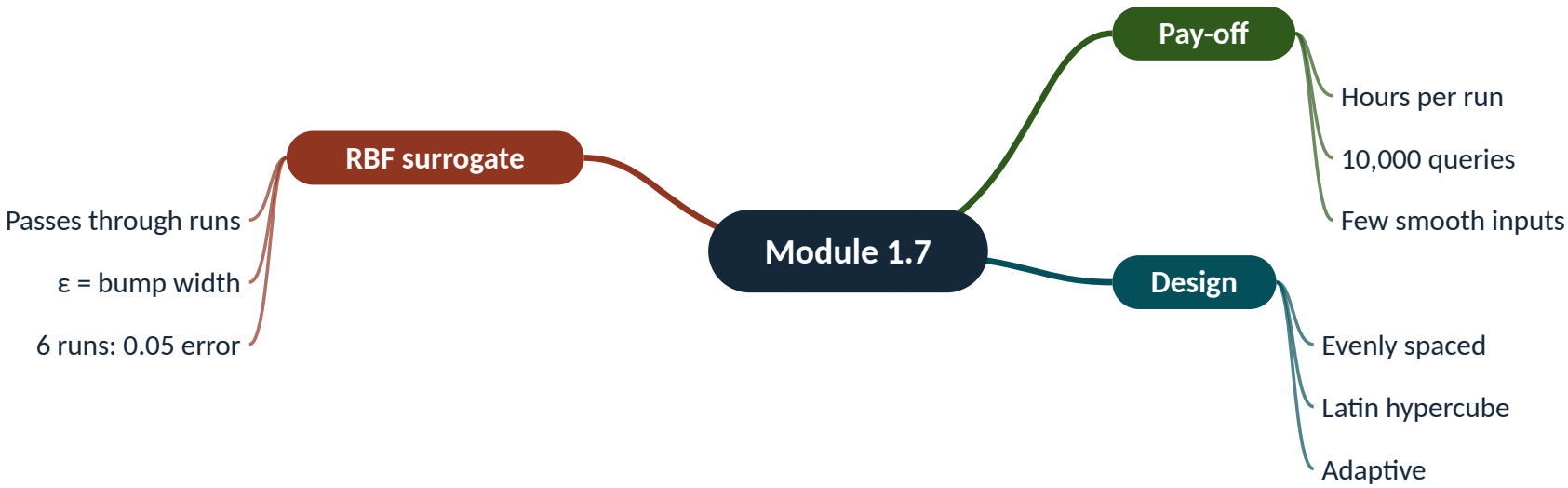
1.5 Physics-informed learning: the core idea



1.6 Data, noise and overfitting



1.7 Surrogates and emulators: fast stand-ins



1.8 Physical AI: sense, estimate, decide, act

